VITAMIN D3

Hypercalcaemia with overdose

Very common deficiency

CHEMICAL NAME (3β,5Z,7E)-9,10-secocholesta-5,7,10(19)-trien-3-ol (cholecalciferol); 1 mcg = 40 IU

CLASS Fat-soluble vitamin D (requires hepatic and renal activation)

TRADE NAMES Vigantol · D-Vit · Fultium-D3 · Devarol (ampoules)

THERAPEUTIC USES / INDICATIONS

- Prevention and treatment of vitamin D deficiency and insufficiency.
- Rickets (children) and osteomalacia (adults).
- Osteoporosis (with calcium and bone therapy).
- Prophylaxis in breastfed infants, pregnancy, elderly, dark skin, limited sun exposure.

ADMINISTRATION

- **PO** daily, weekly or monthly with a fatty meal; drops for infants.
- **IM** depot (high-dose ampoules, e.g. 200,000–600,000 IU) in some countries for malabsorption — less predictable.

DOSAGES

- **Deficiency treatment (adult):** 50,000 IU weekly for 6–8 weeks (or 4,000 IU daily for 10 weeks), then maintenance.
- **Maintenance:** 800–2,000 IU daily.
- **Infants:** 400 IU daily from birth (breastfed).
- **Children (rickets):** 2,000–6,000 IU daily for 2–3 months.

PHARMACOKINETICS

- Absorbed with dietary fat (needs bile).
- Hydroxylated in liver to 25-OH-D (storage form — measured), then in kidney to 1,25-$(OH)_2$D (calcitriol) — active.
- Stored in fat; **t½:** 25-OH-D ~2–3 weeks.

PHARMACODYNAMICS

- Calcitriol acts on **vitamin D receptors** → ↑ intestinal calcium and phosphate absorption, bone mineralisation; regulates PTH; muscle function.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Usually none at standard doses.
- Nausea, constipation (high doses).

ADVERSE EFFECTS (SERIOUS)

- Hypercalcaemia (toxicity): thirst, polyuria, nausea, confusion, weakness, kidney stones, arrhythmias.
- Hypercalciuria; nephrocalcinosis.

CONTRAINDICATIONS & CAUTIONS

- Hypercalcaemia; vitamin D toxicity.
- Metastatic calcification.
- **Caution:** sarcoidosis/granulomatous disease (↑ calcitriol), kidney stones, renal impairment (use active forms).

KEY INTERACTIONS

- **Thiazides:** ↑ hypercalcaemia risk.
- **Digoxin:** arrhythmias if hypercalcaemic.
- **Orlistat, cholestyramine, mineral oil:** ↓ absorption.
- **Phenytoin, carbamazepine, rifampicin:** ↑ vitamin D catabolism.

NURSING MANAGEMENT

- Check calcium (adjusted) before high-dose loading and 1 month after.
- Measure 25-OH vitamin D where indicated (target >50 nmol/L).
- Advise safe sunlight exposure and diet.
- Identify high-risk groups for prophylaxis.

PATIENT EDUCATION

- Take with a meal containing some fat.
- Do not exceed prescribed doses.
- Report excessive thirst, frequent urination, nausea, confusion.
- Safe sun exposure and foods (oily fish, eggs, fortified foods).

KEY POINTS

- Vitamin D3; needs liver and kidney activation.
- Infants 400 IU daily.
- Deficiency: 50,000 IU weekly × 6–8 weeks, then maintenance.
- Toxicity = hypercalcaemia.